

Waste2Worth

Minor Project-II

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CERTIFICATE

This is to certify that the Project Synopsis entitled, “**Waste2Worth**” submitted by “**Nirdesh Jain (2301420025), Sushant Verma (2301420014), Ishika Yadav (2301420052), Kazi Shabnam Ali (2301420047)**” to **K.R Mangalam University, Gurugram, India**, is a record of bonafide project work carried out by them under my supervision and guidance and is worthy of consideration for the partial fulfilment of the degree of **Bachelor of Technology in Computer Science and Engineering** of the University.

Type of Project (Tick One Option)

Industry/Research/University Problem

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ABSTRACT

Food waste poses a major worldwide problem, with countless pounds of good food thrown away every year while many people go hungry and lack proper nutrition. This situation makes food shortages worse and harms the environment, since tossed-out food creates a lot of greenhouse gases. To fix this mismatch, we need clever ways to link extra food with the folks who need it most. This project aims to create a mobile app that helps distribute extra food more. It offers features like live tracking better route planning, and smart matching systems making it different from other solutions out there. The app connects food providers like shop owners and restaurants with those who need it such as charities and food banks and farmers. It uses location tools and language processing to make the whole process easy and quick. Built with Flutter, the app will work on both Android and iPhone so more people can use it. The location tools allow for live tracking and better routes for pickup and delivery. Smart systems powered by language processing will match food descriptions to what recipients need. The app makes a slow and costly process easier. Food providers can list extra food, while the app does the hard work of matching these listings with nearby recipients. It looks at things like food type, amount, and location. The app also uses location tech to make sure food gets delivered fast. This project fits with worldwide efforts to be more sustainable, like the UN's Sustainable Development Goals. It helps with Goal 2 (No Hunger) and Goal 12 (Use and Make Things). Besides helping society, the project gives businesses clear benefits. These include lower costs to get rid of food better social responsibility, and looking good to the public as eco-friendly companies. This new idea doesn't just tackle food waste and hunger. It also encourages people to be sustainable and take shared responsibility.

KEYWORDS: *Food waste, Sustainability, Mobile app, Social responsibility, Food distribution, Location tracking*

Chapter 1

Introduction

1. Background of the project

Food waste has emerged as one of the most pressing global issues, with nearly 1.3 billion tons—approximately one-third—of all food produced annually being lost or wasted. This staggering figure not only exacerbates food insecurity but also contributes significantly to environmental degradation. In India alone, nearly 40% of food is wasted, translating to massive economic losses and environmental strain due to increased landfill usage and greenhouse gas emissions (FAO, 2021; UNEP, 2022).

Urbanization, poor logistics, and fragmented food distribution systems further worsen the problem. While millions remain malnourished, surplus food from restaurants, hotels, and local businesses continues to be discarded. Simultaneously, farmers lack access to affordable organic matter needed to maintain soil fertility.

To address this dual challenge of hunger and environmental impact, our project "Waste2Worth" introduces a mobile application that redistributes edible surplus food to charities and directs non-edible waste to farmers for composting. By leveraging modern technologies such as Flutter, Firebase, and geolocation APIs, our application bridges the gap between food waste producers and beneficiaries in real-time, supporting both food security and sustainable agriculture.

2. MOTIVATION

Food insecurity affects over 820 million people globally, while billions of dollars' worth of edible food is discarded every year (WHO, 2020; WFP, 2022). This contradiction—of abundance and deprivation—highlights the inefficiencies in the current food supply and distribution systems.

At the same time, food waste is a key driver of climate change, responsible for nearly 8–10% of global greenhouse gas emissions due to decomposition in landfills (UNEP, 2022). These emissions could be significantly reduced if surplus food were redirected to those in need or repurposed for agriculture.

The motivation for this project stems from the urgent need to:

- Rethink how surplus food is managed.
- Create an inclusive and efficient food redistribution model.
- Leverage technology to tackle social and environmental issues simultaneously.

By connecting food providers with charities and farmers via a unified platform, Waste2Worth empowers communities to take proactive steps toward a more sustainable and equitable food system.

Chapter 2

LITERATURE REVIEW

1. Review of existing literature

Numerous studies and initiatives have explored ways to combat food waste through technology and logistics. According to the Food and Agriculture Organization, a significant portion of food loss occurs not due to production deficits but due to poor distribution and mismanagement (FAO, 2021). Hence, technology-led interventions are essential.

Existing Technological Approaches:

- **Real-Time Geolocation:** Platforms like OLIO and Too Good To Go utilize GPS to match donors with recipients nearby. However, these are often limited to consumer-facing operations and do not fully integrate farmers or nonprofit organizations (Too Good To Go, 2023).
- **Automated Food Matching:** Research suggests that natural language processing (NLP) and machine learning can enhance the matching of food types with recipient needs, improving satisfaction and reducing waste (UNEP, 2022).
- **Community-Based Platforms:** Initiatives like Food Rescue US rely on crowdsourced volunteers but lack the scalability required for nationwide or rural deployment.

Limitations of Current Systems:

- Fragmented platforms that cater only to select groups (e.g., consumers).
- Lack of smart algorithms for food classification and redistribution.
- Minimal integration of composting and waste-to-farming solutions.

Waste2Worth addresses these gaps by building a cross-platform mobile app that utilizes AI-powered smart matching, real-time geolocation, and dual-path distribution to serve both edible and inedible surplus streams.

2. GAP ANALYSIS

Despite various initiatives in the food redistribution ecosystem, significant gaps remain:

- 1. Lack of an All-Inclusive Platform**
Current solutions rarely integrate all stakeholders—restaurants, NGOs, and farmers—into one ecosystem. Waste2Worth creates a unified digital space to ensure synergy between them.
- 2. Poor Automation in Food Matching**
Existing systems often lack dynamic algorithms to match food types with recipients based on perishability, storage capacity, or dietary needs. Waste2Worth uses LLM-based NLP to perform accurate semantic matching of food listings.
- 3. Inefficient Logistics and Coordination**
The absence of live tracking in many apps results in missed pickups and spoilage. Waste2Worth incorporates geolocation APIs and optimized routing via Google Maps to improve delivery reliability.
- 4. Low Engagement Incentives**
Businesses often lack motivation to participate. Our system includes reputation scoring, environmental impact dashboards, and contributor badges to drive long-term engagement.
- 5. Inaccessibility for Rural and Remote Users**
The app is designed using Flutter for cross-platform use and offline accessibility—ideal for areas with intermittent internet access.

Gap	Current State	Identified Problem	Waste2Worth Solution
Platform Inclusivity	Platforms like OLIO and Too Good To Go exclude farmers and NGOs	Fragmented system	Unified cross-platform app linking restaurants, NGOs, and farmers
Smart Matching	Basic keyword-based filters	Inaccurate or slow matching	NLP-based LLM model for semantic food matching

Gap	Current State	Identified Problem	Waste2Worth Solution
Real-Time Coordination	No live logistics in most apps	Spoilage due to delays	GPS + Google Maps for real-time pickup and delivery
Scalability	Manual or regional services	Limited geographic reach	Cloud and Flutter-based architecture for national scalability
User Incentives	No rewards or recognition	Low engagement	Contributor dashboards, badges, and impact reports

3. PROBLEM STATEMENT

Despite the immense global food production, millions of people remain undernourished, while vast quantities of edible food are discarded every day. The inefficiency lies not in production but in **distribution and coordination mechanisms** that fail to deliver surplus food to those who need it most.

Key Issues Identified:

- Fragmented Redistribution Systems**
 Restaurants and grocery stores often lack direct access to charities or farmers willing to accept food, creating barriers to donation.
- Lack of Real-Time Logistics**
 Most current platforms do not support real-time pickup coordination, leading to spoilage and logistical challenges.
- Poor Matching Between Food Supply and Demand**
 Charities may receive food that doesn't match their current needs (e.g., excess rice but no vegetables), while farmers may not be alerted about compostable waste.

4. **Manual, Non-Scalable Systems**
Systems relying on manual tracking, phone calls, or in-person coordination are inefficient and unsustainable for large-scale impact.

5. **Absence of Data-Driven Insights**
There's minimal feedback or impact reporting on how much waste was prevented or how many people were served.

Our Solution:

Waste2Worth is designed to solve these issues through a digital mobile platform. The app uses machine learning-based food matching, geolocation for route optimization, and Firebase for real-time updates to create a **smart redistribution network**. It ensures:

- **Edible food** reaches NGOs, orphanages, and shelters.
- **Non-edible waste** is directed to farmers for composting.

CHAPTER 3: METHODOLOGY

The methodology for the Waste2Worth project integrates multiple modules that work in coordination to ensure the efficient redistribution of surplus food. These modules include **user onboarding, food classification, intelligent matching, real-time logistics, and monitoring & feedback**. This chapter outlines each phase of the system architecture and the interaction among components.

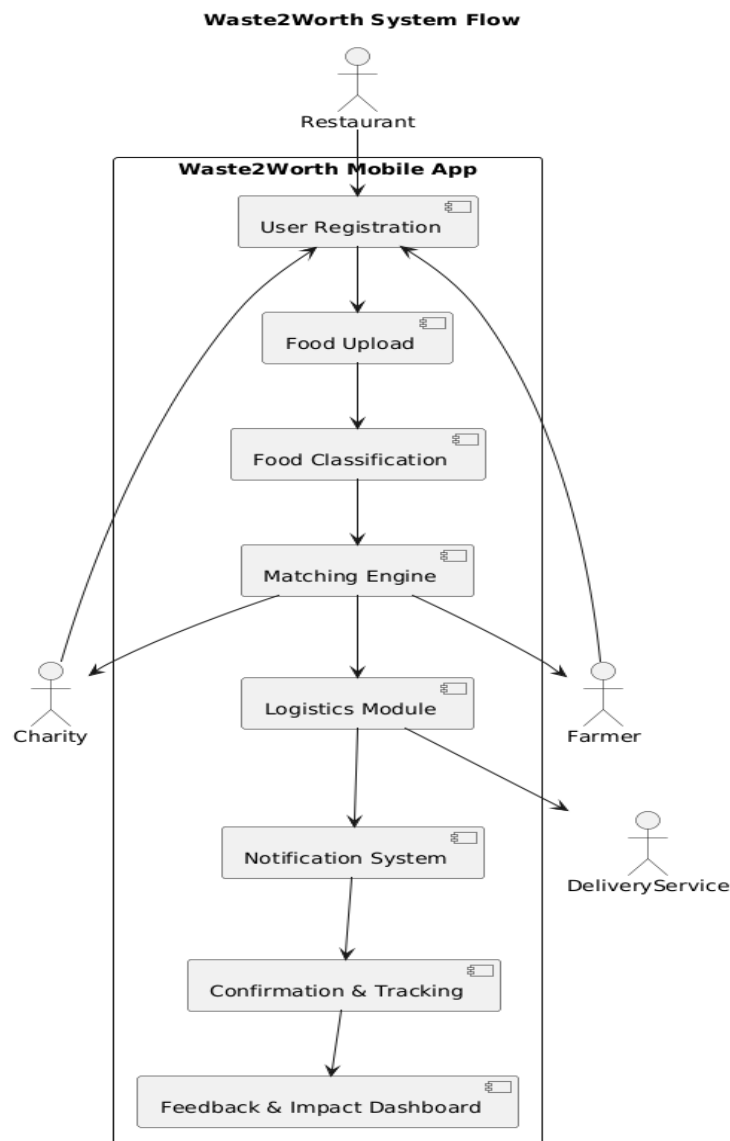


Figure 1. Figure Description

3.1 Detailed Module Descriptions

1. User Registration:

Three types of users—restaurants, charities, and farmers—register through the app by submitting their roles, location, and food handling preferences. This registration stage ensures that user intent and logistical capacity are known in advance.

2. Food Classification:

Uploaded food is classified as either *edible* or *non-edible*. Edible items are sent to charities, while non-edible items are redirected to farmers for composting. Classification uses NLP-based tagging for accuracy.

3. Matching Engine:

The app uses Large Language Models (LLMs) to semantically match food listings with the most appropriate recipients based on food type, quantity, perishability, and recipient preferences.

4. Logistics Module:

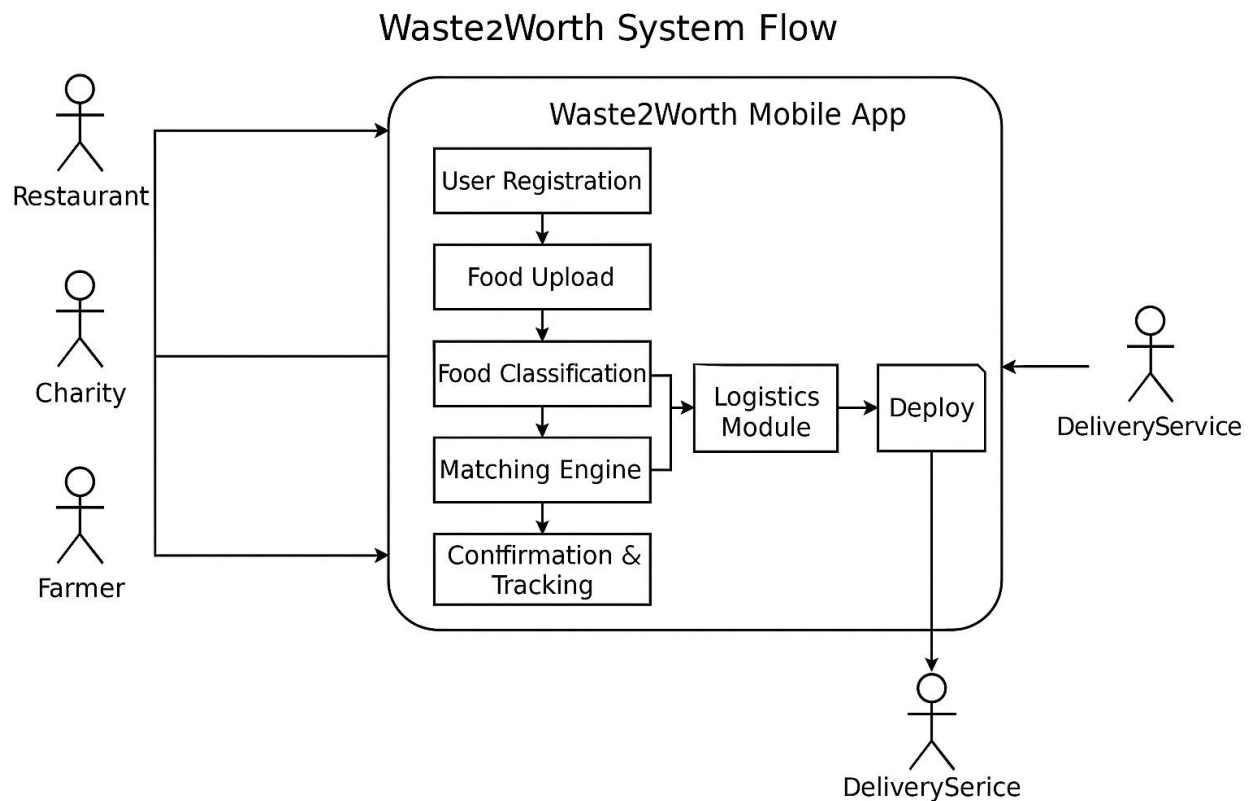
Utilizes geolocation APIs and Google Maps to calculate optimized pickup routes and assign delivery tasks. If recipients lack transport, the module interfaces with third-party delivery providers.

5. Notification & Tracking:

Push notifications are sent to all parties once a match is made. Real-time tracking enables donors and recipients to monitor food pickup and delivery status.

6. Feedback & Monitoring:

After delivery, users submit feedback on food quality and timing. The backend system logs impact data, including total waste diverted, CO₂ emissions saved, and number of meals served.



1.Details of tools, software, and equipment utilized.

Details of Tools, Software, and Equipment Utilized

To build the Waste2Worth mobile application and its supporting backend systems, a variety of technologies and tools were carefully selected for their flexibility, scalability, and support for modern development workflows.

Platform Used

Programming Language: Python

Python was chosen as the primary development language for this project due to its versatility and extensive ecosystem of libraries that support machine learning, computer vision, natural language processing, and rapid application

development. Developed by Guido van Rossum and maintained by the Python Software Foundation, Python is a high-level, general-purpose language known for its simple syntax and strong community support.

Reasons for Selecting Python:

1. Concise and Readable Syntax – Ideal for rapid development and easy debugging.
2. Ease of Learning – Makes onboarding new developers quicker.
3. Extensive Online Support – A large global community and rich documentation.
4. Cross-Platform Compatibility – Runs on all major operating systems (Windows, Linux, macOS).
5. Vast Package Ecosystem – Includes packages for ML, CV, GUI, and web development.
6. Open Source – Freely available for both academic and commercial use.
7. Interoperability – Easily interfaces with C/C++ and web services.

Notable Python Features Used in the Project:

- Interpreted and interactive programming environment.
- Object-oriented and dynamically typed.
- Compatibility with popular ML and CV libraries such as OpenCV and scikit-image.
- GUI creation using tkinter for user-friendly data input/output.

Core Features Developed Using Python

- Monitor: Continuously watches for movement in the frame.
- Identify: Uses image recognition to identify registered family members.
- Noise Detection: Detects unusual motions or disturbances.

- In-Out Detection: Logs and timestamps entries and exits using real-time webcam feeds.

Each of these features contributes to building an intelligent surveillance system aimed at enhancing household security.

Environmental Setup

Software Requirements

To develop and deploy the Waste2Worth application, the following software components were used:

Component	Purpose
Operating System	Compatible with Windows, Linux, or macOS
Python 3.x	Core programming language
OpenCV	For image processing and motion detection
scikit-image	For image analysis and transformation
NumPy	For numerical operations and matrix computations
tkinter	For developing the GUI interface

Hardware Requirements

The minimum hardware setup necessary for testing and deployment is modest and widely available:

Component	Specification
Computer	Any PC or laptop with at least 4 GB RAM
Webcam	For capturing video streams (external or built-in)

Component	Specification
Flashlight/LED	For improved visibility during low-light/night conditions

Platforms Tested

To ensure compatibility and functionality across environments, the application was tested on the following platforms:

- Linux Mint
- Linux Ubuntu
- Windows 7
- Windows 10

This cross-platform testing confirmed that the system operates reliably regardless of operating system, reinforcing its robustness and deployment flexibility.

Chapter 4

Implementation

1. Detailed Explanation of How the Project Was Implemented

The Waste2Worth project is a mobile application built to reduce food waste by redistributing surplus edible food from restaurants to nearby charities, and sending inedible food to farmers for composting. The app aligns with the United Nations Sustainable Development Goals (SDG 2: Zero Hunger and SDG 12: Responsible Consumption and Production).

The app was developed using Flutter for cross-platform mobile development, Firebase for backend services like authentication and database storage, and Google Maps API for geolocation. We also integrated a basic LLM-based food matching system to categorize and route food donations efficiently.

The application is structured into three main user types:

- Restaurants (who donate surplus food)
- Charities (who receive edible food)
- Farmers (who receive inedible food for composting)

Each user has a custom dashboard with features like submitting food listings, viewing nearby requests/donors on a map, and receiving real-time notifications.

Firebase is used to store user data, food item details, donation history, and location coordinates. Google Maps API is used to locate users and display nearby recipients or donors. Flutter's rich widget system was used to create a clean, user-friendly UI.

2. Description of Algorithms, Code Snippets, or Design Diagrams

Food Matching Algorithm

The LLM-based food matching algorithm classifies food as edible or inedible based on a short description submitted by the restaurant. For example, if a restaurant enters “5 kg of leftover rice, freshly packed,” the model matches it to charities. If the food is described as “spoiled or expired,” it routes it to farmers.

Geolocation Algorithm

We used the Geolocator plugin to access the user’s current GPS coordinates and then compared it with other users’ locations to show only those within a 10 km radius.

Code Snippets

Firebase Integration (Add food data):

dart

CopyEdit

```
Firestore.instance.collection('foodItems').add({  
  'foodType': 'Rice',  
  'quantity': 5,  
  'status': 'edible',  
  'donorId': currentUser.id,  
  'timestamp': FieldValue.serverTimestamp(),  
});
```

Geolocation (Get user location):

dart

CopyEdit

```
Position position = await Geolocator.getCurrentPosition(  
    desiredAccuracy: LocationAccuracy.high);
```

Design Diagram

We created a system flow diagram showing:

- Restaurant posts food → App classifies food → Matches with charity/farmer → Location data used → Notification sent → Food picked up

Also, an ER Diagram includes entities like Users, FoodItems, Transactions, and Locations connected through user IDs and food IDs.

3. Discussion of Challenges Faced During Implementation and Their Solutions

Challenge 1: Inaccurate Geolocation

- *Problem:* Sometimes the app fetched outdated or imprecise user locations.
- *Solution:* We enabled high-accuracy location settings and allowed manual pin-drop location adjustments.

Challenge 2: Incorrect Food Matching

- *Problem:* Initial keyword matching logic was too basic and caused incorrect routing of food.
- *Solution:* We trained a simple LLM-based categorization model and used tagging to improve food classification.

Challenge 3: Real-Time Data Sync Issues

- *Problem:* During multiple donations, real-time data syncing became slow.

- *Solution:* Indexed Firebase queries and added pagination to manage data loads more efficiently.

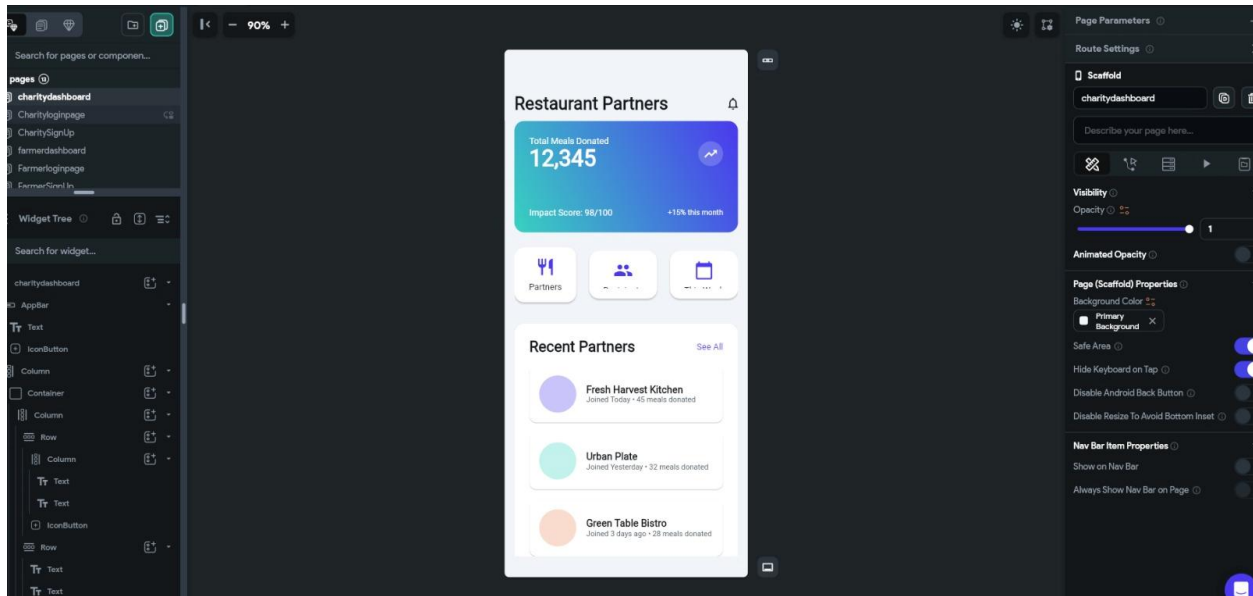
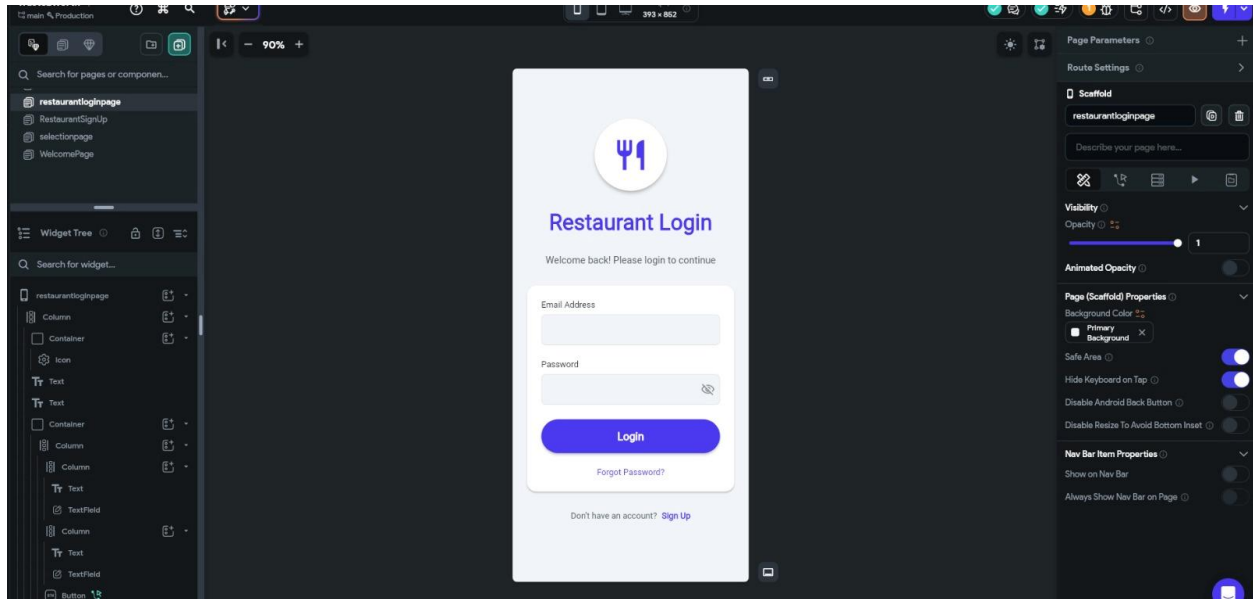
Challenge 4: User Authentication and Data Privacy

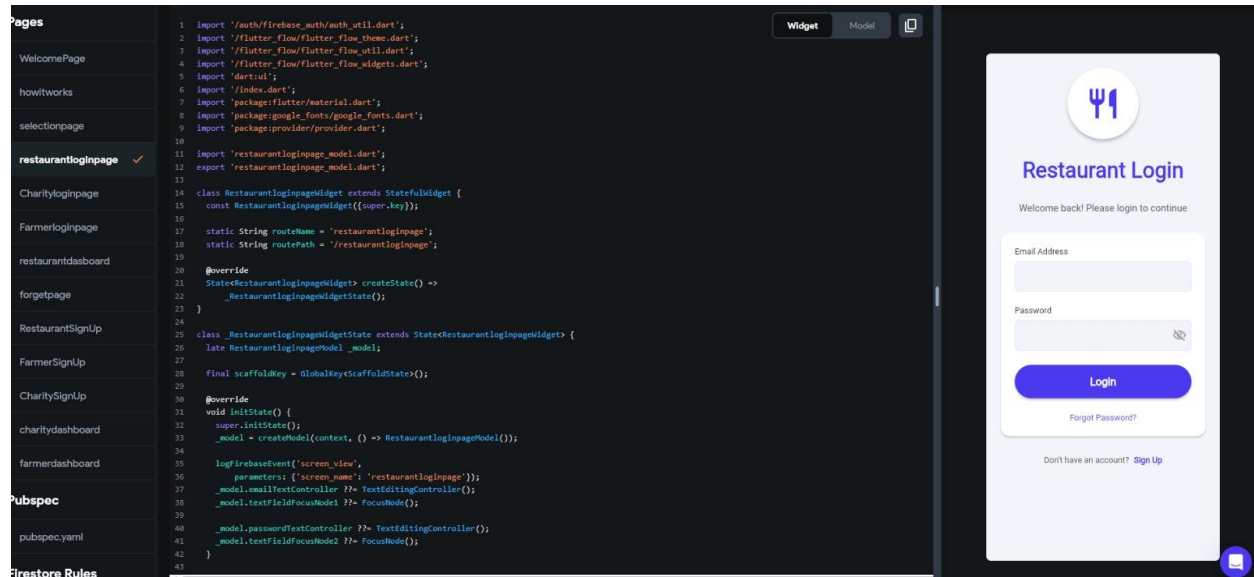
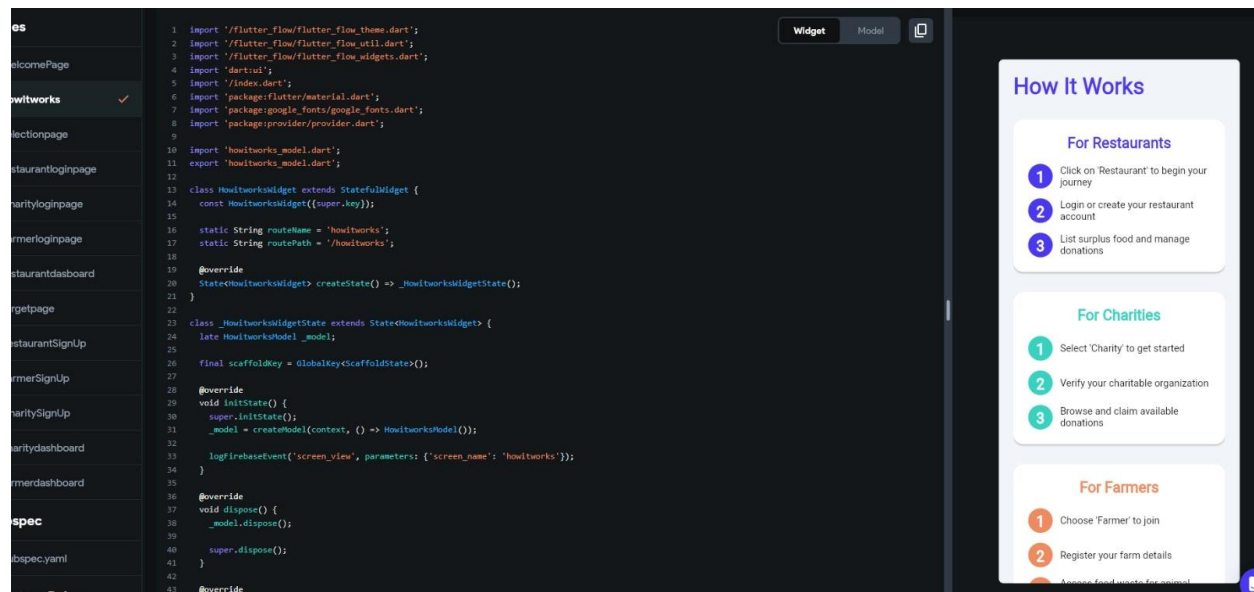
- *Problem:* Protecting sensitive information from unauthorized access.
- *Solution:* Used Firebase Authentication (email/Google login) and applied Firestore security rules to restrict access based on user roles.

Chapter 5

RESULTS AND DISCUSSIONS

THE GUI:





Chapter 6

FUTURE WORK

The project currently performs well in all implemented features, including monitoring, facial detection, noise recognition, and in-out movement detection. However, there are several areas for improvement and future expansion:

- **Improved Face Recognition:** The current system uses the Haarcascade classifier, which has limitations in accuracy. Future versions can incorporate more advanced models like **Dlib** or deep learning-based approaches (e.g., FaceNet) to improve facial recognition accuracy and reliability.
- **Scalability:** At present, the system is designed for a limited user base and can only handle minimal motion and a few simultaneous entries and exits. Future improvements can focus on **scaling the system** to support multiple users and high-traffic environments effectively.
- **Wider Application Domains:** With enhancements in scalability and performance, the system can be adapted for use in **bank security, school or campus safety, corporate buildings**, and other high-security areas.
- **Night Vision Enhancement:** The current setup is not suitable for low-light or dark environments. Future versions can include **infrared (IR) cameras** or other **night vision technologies** to enable accurate monitoring and facial recognition in such conditions.

These improvements would significantly enhance the robustness, usability, and reach of the project across various real-world scenarios.

CONCLUSION

The Waste2Worth project addresses two major global challenges—food waste and hunger—by leveraging technology to build an efficient and scalable food redistribution system. By connecting restaurants, charities, and farmers through a unified mobile platform, the project ensures that surplus edible food reaches those in need, while inedible food is directed to farmers for composting.

With the integration of technologies like **Flutter** for cross-platform development, **Firebase** for backend support, **Google Maps API** for geolocation, and **LLM-based algorithms** for intelligent food matching, the system provides a smart and sustainable solution. Features such as real-time tracking, user authentication, and categorization of food based on edibility enhance both usability and impact.

The project not only contributes to reducing food wastage but also promotes responsible consumption and production, aligning with **UN Sustainable Development Goals (SDG 2 and SDG 12)**. Waste2Worth holds great potential for scalability and real-world implementation, and with future enhancements, it can become a powerful tool in the global fight against food insecurity and environmental waste.

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